

Elysia Darbourne

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TECHNICAL SKILLS

Languages: C#, C++, Python, MATLAB, C, Java, JavaScript, SQL, HTML, CSS, PowerShell, TypeScript, GraphQL

Frameworks: React, ASP.NET Core, .NET Core, Unity

Tools & Technologies: Git, Kubernetes, gRPC, Docker, Pandas, Blender, REST API

WORK EXPERIENCES

Red Bull Racing and Technology

Jul. 2024 - Aug. 2025

F1 Software Engineer Intern

Milton Keynes, UK

- Leveraged C++, Python, MATLAB, Simulink, and real-time operating principles to design simulation systems.
- Worked on wind tunnel applications for analysis of data points with GraphQL.
- Developed software tools to streamline wind tunnel data integration, improving efficiency for engineers.
- Worked with performance engineers to deliver high-fidelity tools to improve car setup and strategy testing.
- Replaced outdated manual processing of data with an automated .NET app, reducing time by 200%.
- Built real-time data analysis applications with Streamlit, Python, and Pandas to visualise telemetry data.
- Implemented a gRPC service to enable fast and reliable service-to-service communication in a distributed system.

ZeMind Game Studio Ltd

Jan. 2024 - Apr. 2024

Game Developer

Toronto, ON

- Developed gameplay features using Three.js for a promotional game supporting a band's album release.
- Worked on a cross-platform Unity game, fixing bugs on iOS devices to ensure robust gameplay.
- Participated in code-reviews, testing, and debugging, maintaining high-quality builds and game stability.

Titanium Agency

Sept. 2022 – Dec. 2022, May. 2023 – Aug. 2023

Software Engineer Intern

Waterloo, ON

- Created revenue prediction algorithm, alerting managers to opportunities in real-time, boosting profitability.
- Implemented 10 new features to improve application capabilities in a central .NET app.
- Designed data transfer applications using GCP, using BigQuery for fast data migration.
- Automated creation of campaigns using Python and APIs, reducing manual effort and accelerating deployment.

PROJECTS

Dungeon Game

C#, Unity

- Created captivating dungeon exploration mechanics, delivering an engaging single-player experience with immersive storytelling.
- Integrated AI-driven enemy behaviours, designing formidable dungeon creatures and encounters to deliver exciting and strategic game play challenges.
- Optimized game performance and loading times, ensuring smooth game play and minimal interruptions during dungeon exploration.
- Crafted intricate level designs, ensuring a seamless flow of exploration and maintaining a balance between difficulty and player progression.

Premier League Connections

Typescript, React

- Created a React app where people get daily puzzles with the challenge of grouping the premier league players by common club.
- Deployed the typescript react app using GitHub pages so that players worldwide can play.
- Utilized typescript to create a robust code base and created a bug-free experience for users.

EDUCATION

University of Waterloo

Bachelor of Computer Science (GPA: 3.9)

2027

Coursework: Data Structures, Algorithms, Operating Systems, Concurrency, Statistics, Combinatorics, Linear Algebra